

IMAGE PROCESSING FOR AGRICULTURE FIELD SEGMENTATION

**A Project Report Submitted in Fulfillment of the Requirements for the
Award of Degree**

BACHELOR OF TECHNOLOGY

In

ELECTRONICS AND COMMUNICATION ENGINEERING

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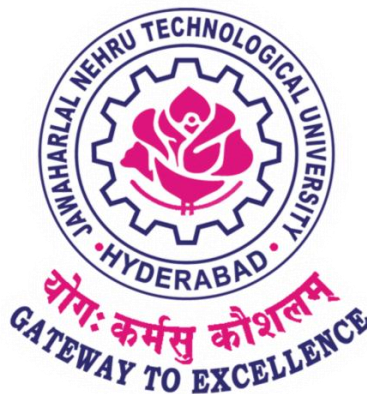
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In Partial Fulfillment of the Requirements for the Degree of
ELECTRONICS AND COMMUNICATION ENGINEERING



CERTIFICATE BY THE HEAD OF THE DEPARTMENT

This is to certify that the Project Report entitled “**IMAGE PROCESSING FOR AGRICULTURE FIELD SEGMENTATION**” being submitted by **K. Yamini, KBS Karthikeya Kasyap, Kavya, Sankeertana** in partial fulfillment for the award of the Degree of **Bachelor of Technology in Electronics and Communication Engineering** at Jawaharlal Nehru Technological University is a record of bonafide work carried out by them. The result embodied in this project report has not been submitted to any other University or Institute for the award of any Degree or Diploma.

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DECLARATION OF THE CANDIDATE

We hereby declare the mini project entitled “**IMAGE PROCESSING FOR AGRICULTURE FIELD SEGMENTATION**” is a bona-fide record work done and submitted under the esteemed guidance of **Dr. T. Satya Savithri**, Professor, Department of ECE, JNTU, Hyderabad, in particular fulfillment of the requirements for the award of the degree of **Bachelor of Technology (IDP)** in **Electronics and Communication Engineering** during the academic year 2025-2026. This record is a bona-fide work carried out by us and the results kept in the mini project have not been reproduced or copied. The results in the mini project have not been submitted in any other university for the award degree or diploma.

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INDEX

Chapter No.	Title	Page No.
1.	INTRODUCTION	1-3
1.1	Motivation of the Project	2
1.2	Objectives	3
2	LITERATURE SURVEY	4-6
3	FUNDAMENTALS OF IMAGE PROCESSING	7-15
3.1	Digital Image	7-8
3.2	Image Preprocessing	9-10
3.3	Feature Extraction	10
3.4	Image Segmentation	10
3.5	Machine Learning	11
3.6	Support Vector Machine	11
3.7	Principal Component Analysis (PCA)	12
3.8	Fixed-Point Representation	12
3.9	Image Memory Representation	12
3.10	Python	12-14
3.11	Verilog HDL	14
3.12	FPGA	15
3.13	Vivado Simulation	15
4	Methodology	16-21
4.1	Overall System Workflow	16-17

4.2	Agriculture Image Acquisition System	18
4.3	Image Preprocessing	18
4.4	Crop Classification using Machine Learning	19
4.5	PCA-Based Visualization	20
4.6	Agricultural Field Segmentation and Crop Percentage Estimation	20
4.7	FPGA-Oriented Data Conversion	21
4.8	FPGA Simulation using Vivado	21-22
5	RESULTS	23-33
5.1	Dataset Processing	23
5.2	PCA Visualization of Feature Distribution	23-24
5.3	Crop Classification and Yield Estimation Results	24-32
	5.3.1 Mirchi	
	5.3.2 Cotton	
	5.3.4 Paddy	
	5.3.5 Sunflower	
5.4	Software vs FPGA Comparison	33
5.5	RTL Schematic of FPGA-Based Crop Detection and Crop Percentage Estimation System	33
	CONCLUSION	34
	FUTURE SCOPE	34
	REFERENCES	35

List of Figures

Fig. No.	Title	Page No.
Fig. 4.1	Overall System Workflow	16-17
Fig. 5.1	PCA Visualization of Feature Distribution	24
Fig.5.2.1	Original vs Binary Mask (Mirchi)	25
Fig. 5.2.2	Python Simulation Output (Mirchi)	25
Fig. 5.2.3	Vivado Simulation (Mirchi)	26
Fig. 5.2.4	Simulation outputs (Mirchi)	26
Fig.5.3.1	Original vs Binary Mask (Cotton)	27
Fig. 5.3.2	Python Simulation Output (Cotton)	27
Fig. 5.3.3	Vivado Simulation (Cotton)	28
Fig. 5.3.4	Simulation outputs (Cotton)	28
Fig.5.4.1	Original vs Binary Mask (Paddy)	29
Fig. 5.4.2	Python Simulation Output (Paddy)	29
Fig. 5.4.3	Vivado Simulation (Paddy)	30
Fig. 5.4.4	Simulation outputs (Paddy)	30
Fig.5.5.1	Original vs Binary Mask (Sunflower)	31
Fig. 5.5.2	Python Simulation Output (Sunflower)	31
Fig. 5.5.3	Vivado Simulation (Sunflower)	32
Fig. 5.5.4	Simulation outputs (Sunflower)	32
Fig 5.7	RTL Schematic of FPGA-Based Crop Detection and Crop Percentage Estimation System	33

List of Tables

Table No.	Title	Page No.
Fig. 5.6	Software vs FPGA Comparison	33

ABSTRACT

Agriculture has emerged as a vital field for improving crop productivity, resource optimization, and automated field monitoring through advanced imaging and computational techniques. Within this domain, image-based crop analysis plays a significant role in identifying crop types, estimating crop coverage, and supporting data-driven decision making for sustainable farming. However, farmers and agricultural systems often face challenges in accurately distinguishing multiple crops present in the same field image, leading to difficulties in monitoring crop growth, identifying dominance among crops and planning field operations effectively.

This project addresses these issues by developing a multi-crop image segmentation system capable of identifying and estimating the area covered by specific crops such as sunflower, paddy, cotton, and mirchi within composite images. The system first classifies the crops present using a Support Vector Machine (SVM) model trained on a curated dataset of uniformly resized images. Once the target crop, such as sunflower is detected, colour-based thresholding techniques are applied to segment the crop region from the background and other crops. The segmented mask is then used to compute the exact area occupied by the crop as well as the percentage of coverage within the image.

After finding the crop yield percentage using image processing, the input pixel values were extracted and used to calculate the weights and bias values for the model. These values were then implemented in the Vivado Design Suite software to estimate the yield percentage through hardware implementation. This process helps in achieving faster processing and efficient analysis for crop yield prediction.

CHAPTER 1

INTRODUCTION

Agriculture is one of the most important sectors for food production and economic development. In countries like India, agriculture supports the livelihood of millions of people and contributes significantly to the national economy. With the increasing population and food demand, there is a strong need for advanced agricultural technologies that improve crop productivity, reduce crop losses, and support efficient field monitoring systems.

Traditional agricultural monitoring methods mainly depend on manual observation and field inspection. Farmers visually inspect agricultural fields to identify crop conditions, estimate crop area, and monitor field productivity. However, these methods require more time, labor, and human effort. Manual estimation may also produce inaccurate results, especially in large agricultural fields containing multiple crops. Environmental conditions such as lighting variations, shadows, soil conditions, and mixed vegetation further increase the difficulty of crop monitoring.

Recent advancements in digital image processing, machine learning, and hardware acceleration technologies have enabled the development of automated agricultural monitoring systems. Image processing techniques help analyze agricultural images, identify crop regions, estimate crop area, and improve farming efficiency. Machine learning algorithms such as Support Vector Machine (SVM) provide accurate crop classification by learning patterns from training datasets.

The proposed project focuses on “Image Processing for Agriculture Field Segmentation using SVM and Vivado Simulation.” The system uses sample crop images such as Paddy, Cotton, Sunflower, and Mirchi for training the SVM classifier. The input images are preprocessed using resizing and RGB conversion techniques. The processed images are converted into feature vectors and provided to the SVM classifier for training.

After training, a test image is given as input to the system. The trained SVM classifier predicts the crop type present in the image. Once the crop is identified, RGB-based segmentation is performed to isolate crop pixels from the background. Different crops possess unique RGB intensity characteristics, and threshold conditions are used to identify crop regions accurately.

The segmented crop pixels are counted to estimate crop coverage percentage. The crop percentage is calculated using the ratio between detected crop pixels and total image pixels. The

trained SVM model generates weights and bias values, which are exported into hexadecimal fixed-point representation and implemented using Vivado FPGA.

1.1 Motivation of the Project:

The increasing demand for food production and efficient farming methods has created a strong need for intelligent agricultural monitoring systems. Traditional farming methods mainly depend on manual crop observation and field inspection, which require more labor and time. In large agricultural fields, manual monitoring becomes difficult and may produce inaccurate results.

Modern agricultural systems require automatic crop monitoring techniques that can identify crops accurately and estimate crop area efficiently. Image processing and machine learning technologies provide efficient solutions for automated agricultural analysis.

The proposed project is motivated by the following factors:

- Precision agriculture helps improve crop productivity and reduce resource wastage. Accurate crop monitoring systems are essential for efficient farming.
- Different crops possess different characteristics. Automatic crop classification helps farmers identify crop distribution and monitor agricultural fields efficiently.
- Manual crop area estimation is difficult and time-consuming. Automated image-processing systems provide accurate crop percentage estimation.
- Real-time agricultural monitoring systems are required for smart farming applications. Faster image processing improves agricultural decision making.
- Machine learning algorithms such as SVM provide high classification accuracy for agricultural image analysis.
- FPGA implementation improves processing speed and reduces execution delay. FPGA systems support parallel processing and real-time image analysis.
- Different crops possess unique RGB intensity characteristics. RGB thresholding provides efficient crop segmentation and crop area estimation.
- The project supports smart farming systems, drone-based agricultural monitoring, and automated yield estimation systems.

Thus, the project helps improve agricultural monitoring efficiency using image processing, machine learning, and FPGA technologies.

1.2 Objectives:

The main objective of this project is to develop an efficient crop classification and crop percentage estimation system for agricultural monitoring applications using image processing, Support Vector Machine (SVM), and FPGA oriented implementation techniques.

The project focuses on improving crop analysis accuracy and achieving real-time processing performance using hardware based implementation methods. It includes:

- To collect and preprocess agricultural crop images.
- To extract features from crop images and classify them using SVM classifiers.
- To detect the target crop (e.g., sunflower) in an image containing multiple crops.
- To apply threshold-based segmentation on identified crop regions for area extraction.
- To compute the percentage of the target crop with respect to the entire image area.
- To Simulate algorithms in Python and Vivado.

CHAPTER 2

LITERATURE REVIEW

Digital image processing forms the fundamental basis for modern agricultural monitoring systems. Gonzalez and Woods [1], in *Digital Image Processing*, presented essential image processing techniques such as image enhancement, segmentation, filtering, feature extraction, and classification. Their work provides the theoretical foundation required for designing image-based crop detection systems. In our project, these concepts are directly applied for preprocessing agricultural images, isolating crop regions, and extracting meaningful information for crop classification and percentage estimation.

Patil and Bodhe [2] developed a plant leaf disease detection system using image processing techniques. Their methodology involved image acquisition, preprocessing, segmentation, feature extraction, and classification for identifying diseased plant regions. This work demonstrates how automated image analysis can reduce manual agricultural monitoring efforts. Similarly, our project uses image preprocessing and segmentation techniques to identify crop regions, but instead of disease detection, the focus is on crop detection and estimation of crop coverage percentage.

Revathi and Hemalatha [3] proposed a cotton leaf spot disease detection system using digital image processing methods. Their work emphasized accurate segmentation and pattern recognition to identify infected regions in plant leaves. The importance of segmentation shown in this work directly relates to our project, where segmentation is used to separate crop regions from the background and distinguish multiple crop types within an agricultural image.

Chaki and Parekh [4] presented a plant image segmentation approach based on colour and texture features. Their study highlighted that combining colour and texture analysis improves segmentation accuracy in plant images. Since agricultural images often contain complex backgrounds and varying lighting conditions, this work is highly relevant to our project, where accurate segmentation is necessary for crop classification and percentage calculation.

Al-yoonus and Al-Kazzaz [5] implemented a basic video processing system on FPGA for real-time applications and analyzed timing performance using FPGA tools. Their work demonstrated that FPGA architectures significantly improve processing speed compared to software-based systems. This directly supports our project, which follows an FPGA-oriented implementation approach. Although our current work is limited to simulation in Vivado rather than physical FPGA deployment, this reference validates the feasibility of future real-time hardware implementation.

Arivazhagan et al. [6] proposed a texture-based approach for detecting unhealthy plant leaf regions and classifying plant diseases. Their work showed that texture features are highly effective in distinguishing different plant conditions. Feature extraction is similarly important in our project, where visual characteristics such as texture, shape, and color can be used for crop identification and classification.

Amara et al. [7] introduced a deep learning-based approach for banana leaf disease classification. Their work demonstrated that machine learning techniques can significantly improve classification accuracy compared to conventional methods. Although our project uses traditional machine learning techniques rather than deep learning, this work highlights the importance of intelligent classification algorithms for agricultural image analysis and supports the use of automated crop recognition systems.

Sladojevic et al. [8] developed a deep neural network model for recognizing plant diseases from leaf images. Their system achieved high classification performance by automatically learning image features instead of relying on manually extracted features. This study is relevant to our work because it demonstrates the effectiveness of AI-based agricultural monitoring systems. While our project focuses on crop detection and percentage estimation, the concept of automated visual classification remains similar.

Senthilkumaran and Rajesh [9] surveyed various edge detection techniques used in image segmentation. Their work emphasized the importance of boundary detection for separating objects from the background in digital images. Edge detection methods can support our project during crop segmentation, especially in identifying clear crop boundaries required for accurate area estimation.

Jain [10], in *Fundamentals of Digital Image Processing*, explained the principles of image representation, filtering, enhancement, segmentation, and feature extraction. This work provides an additional theoretical foundation for our project, especially in designing preprocessing and segmentation stages for agricultural image analysis.

From the reviewed literature, it is clear that image processing and machine learning techniques are widely used for agricultural applications such as disease detection, segmentation, and classification. Several studies demonstrate the effectiveness of automated crop image analysis, while FPGA-based research highlights the potential for real-time implementation. However, most existing works focus either on disease detection, segmentation, or hardware acceleration individually. Limited research integrates crop detection, crop percentage estimation, machine learning, and FPGA-oriented implementation into a single framework. Therefore, our project aims to bridge this gap by developing an agricultural monitoring system capable of detecting crops, estimating crop coverage percentage, and evaluating FPGA-based implementation through Vivado simulation for future real-time deployment.

CHAPTER 3

IMAGE PROCESSING FUNDAMENTALS

3.1 Digital Image

Digital image fundamentals describe the basic concepts used in image processing systems for agricultural analysis and crop monitoring. A digital image is a visual representation of an agricultural field stored electronically as a collection of pixels arranged in rows and columns. Each pixel contains numerical information representing brightness or colour intensity at a specific location.

In agricultural image processing, digital images are analysed to identify crop regions, classify crop types, and estimate crop coverage percentage. Since all image processing operations depend on pixel analysis, understanding digital image fundamentals is essential for designing efficient agricultural monitoring systems.

1. Pixels

A pixel is the smallest unit of a digital image. The term pixel stands for “picture element.” Every digital agricultural image consists of thousands of pixels arranged in rows and columns.

Each pixel stores information about brightness or color. In crop monitoring applications, pixel values help distinguish crops from soil, background, or unwanted objects.

For example:

- Green pixels may indicate paddy crops
- White pixels may indicate cotton
- Yellow pixels may indicate sunflower

Thus, crop analysis depends heavily on pixel-level operations.

2. Resolution

Resolution refers to the total number of pixels present in an image and determines image clarity and detail. It is represented as:

Width × Height

Examples:

- 64 × 64

Higher resolution provides better crop detail but increases memory usage and processing complexity. In FPGA-oriented systems, moderate resolution is preferred to balance accuracy and speed.

3. RGB Images

RGB images are colour images formed using red, green, and blue channels.

Each pixel contains three intensity values:

- Red
- Green
- Blue

Agricultural field analysis depends strongly on colour because different crops show unique colour distributions.

Examples:

- Paddy → green dominant
- Cotton → white dominant
- Sunflower → yellow dominant
- Mirchi → red/green dominant

RGB images provide more information than grayscale images, making them highly suitable for crop classification.

4. Image Matrix Representation

A digital image is stored as a matrix.

For grayscale:

$$I(x, y)$$

For RGB:

$$I(x, y, 3)$$

Example:

A 64×64 RGB image: $64 \times 64 \times 3$

This matrix-based representation is important because image processing algorithms operate mathematically on pixel matrices.

3.2 Image Preprocessing

Image preprocessing improves image quality and prepares data for analysis.

Raw agricultural images may contain:

- varying sizes
- noise
- brightness variations
- inconsistent colour distribution

Preprocessing helps standardize images before classification and segmentation.

1. Image Resizing

Image resizing is an essential preprocessing step that converts all input images to a fixed dimension for uniform analysis. Since agricultural images may have different sizes and resolutions, resizing ensures consistent feature extraction and proper machine learning classification. In this project, images are resized to 64×64 pixels to reduce computational complexity, improve processing speed, and simplify FPGA-oriented implementation through fixed memory and hardware requirements.

2. Colour Space Conversion

Colour space conversion is an important preprocessing step that converts images from one color format to another for efficient analysis. Different image processing applications use different colour spaces depending on the required information. In this project, images are converted from BGR to RGB format because OpenCV loads images in BGR format by default, while RGB representation is more suitable for agricultural analysis. Since crop classification and field segmentation depend on colour characteristics, RGB helps accurately distinguish crops based on their red, green, and blue intensity values.

3. Normalization

Normalization is a preprocessing technique used to scale pixel intensity values into a fixed range for efficient machine learning processing. In digital images, pixel values normally range from 0 to 255, and normalization converts them to a smaller range of 0 to 1 by dividing each value by 255. This reduces numerical variation, improves classifier stability, and enables faster and more accurate learning during crop classification.

3.3 Feature Extraction

Feature extraction is the process of converting image data into numerical features that can be used by machine learning algorithms for classification. Instead of analysing images visually, the system represents important image characteristics such as colour, texture, shape, and intensity in numerical form. In this project, RGB pixel values are directly used as features, where each 64×64 RGB image is flattened into a 12,288-dimensional feature vector. This structured representation enables the SVM classifier to efficiently learn and distinguish different crop types.

3.4 Image Segmentation

Image segmentation is an important image processing technique used to divide an image into meaningful regions for easier analysis. In agricultural applications, segmentation helps separate crop regions from soil, background, and other unwanted areas, improving the accuracy of crop identification and area estimation. Different segmentation methods are available, but colour-based segmentation is highly effective for agricultural images because different crops exhibit distinct colour characteristics, making region separation simpler and more accurate.

Thresholding

Thresholding is a simple image segmentation technique used to classify pixels based on predefined intensity or color conditions. It compares pixel values with specific threshold limits to determine whether a pixel belongs to the target region or the background. In agricultural field analysis, thresholding is effective because crops often have distinct colour characteristics. For example, paddy crops contain green-dominant pixels, so conditions based on RGB intensity can be used for detection. Thresholding is widely preferred because it is fast, computationally simple, and suitable for FPGA-oriented implementation.

3.5 Machine Learning

Machine learning is an intelligent computational approach that enables automatic crop classification by learning patterns from training data instead of relying on manually defined rules. It analyzes image features such as color, texture, and intensity to identify differences between crop types and make accurate predictions for new input images. In agricultural applications, machine learning improves automation, reduces human effort, and supports tasks such as crop detection, disease identification, yield estimation, and plant recognition.

3.6 Support Vector Machine Fundamentals

Support Vector Machine (SVM) is a supervised machine learning algorithm widely used for classification tasks due to its high accuracy and efficient decision-making capability. SVM works by finding an optimal separating boundary called a hyperplane that maximizes the margin between different classes, improving classification performance. For binary classification, the decision function is expressed as:

$$f(x)=w^Tx+b$$

where:

(x) = input feature vector

(w) = weight vector

(b) = bias term

The classification decision is:

$$f(x)>0\rightarrow\text{Class 1}$$

$$f(x)<0\rightarrow\text{Class 2}$$

In this project, since multiple crop classes are involved, One-vs-One (OvO) multiclass classification is used, where separate binary classifiers are created for each pair of classes. The number of classifiers required is:

$$n(n-1)/2$$

For 4 crop classes:

$$\frac{4(4-1)}{2} = 6$$

Thus, six binary SVM classifiers are used for crop classification. SVM is suitable for this project because of its simple mathematical implementation, efficient performance with smaller datasets, and FPGA-friendly arithmetic operations.

3.7 Principal Component Analysis (PCA)

Principal Component Analysis (PCA) is a dimensionality reduction technique used to simplify high-dimensional image data while preserving the most important information. In image processing applications, feature vectors may contain thousands of dimensions, making visualization and analysis difficult. PCA transforms the original data into a smaller set of principal components that capture the maximum variance in the dataset. This reduces computational complexity and helps visualize crop class distribution and classifier boundaries more effectively.

3.8 Fixed-Point Representation

Fixed-point representation is commonly used in FPGA-based systems because it provides faster and more efficient arithmetic processing compared to floating-point representation. Floating-point operations require complex hardware, higher memory usage, and increased processing time, making them less suitable for real-time FPGA applications. In fixed-point representation, decimal values are converted into scaled integer values, which simplifies arithmetic operations such as multiplication and addition. This reduces hardware complexity, improves processing speed, and makes FPGA implementation more resource efficient.

3.9 Image Memory Representation

Image memory representation refers to the method of storing digital image data in hardware memory for efficient processing. In FPGA-based systems, images are stored in a structured format using techniques such as row-major storage, pixel indexing, and memory addressing to allow fast access during computation. For RGB images, each pixel is represented using 24 bits, consisting of 8 bits each for red, green, and blue components. Proper memory representation is essential for FPGA simulation because image processing operations depend on accurate and efficient retrieval of pixel data from memory.

3.10 Python

Python is a high-level programming language widely used for image processing, machine learning, and scientific computing because of its simple syntax, extensive libraries, and efficient data handling capabilities. In the proposed project, Python is used for agricultural image preprocessing, feature extraction, crop classification, segmentation, visualization, and FPGA

data preparation. Python enables rapid development and easy integration of image processing and machine learning algorithms.

1. OpenCV

OpenCV (Open Source Computer Vision Library) is a powerful Python library widely used for image processing and computer vision applications due to its fast and efficient image handling capabilities. It provides built-in functions for image acquisition, resizing, colour space conversion, filtering, and pixel-level analysis. In this project, OpenCV is used for reading agricultural images, resizing them to a fixed resolution, converting images from BGR to RGB format, and performing preprocessing operations required for crop classification and segmentation. Its efficient processing makes it highly suitable for agricultural image analysis applications.

2. NumPy

NumPy is a powerful numerical computing library in Python used for efficient array processing and mathematical computations. Since digital images are represented as multidimensional arrays of pixel values, NumPy plays an important role in image processing applications. In this project, it is used for pixel manipulation, feature extraction, normalization, matrix operations, and fixed-point data conversion for FPGA compatibility. Its optimized array operations improve computational speed and overall efficiency in machine learning and image processing tasks.

3. Matplotlib

Matplotlib is a Python visualization library used for plotting graphs, displaying images, and analysing data visually. In image processing and machine learning applications, visualization helps understand system performance and verify results effectively. In this project, Matplotlib is used to display original agricultural images, visualize binary segmentation masks, plot PCA-based crop distribution, and analyze SVM classification boundaries. This helps evaluate classification accuracy and segmentation performance clearly.

4. Scikit-Learn

Scikit-Learn is a Python machine learning library widely used for implementing classification, data analysis, and dimensionality reduction algorithms. It provides efficient built-in models and tools that simplify machine learning development without requiring complex implementation from scratch. In this project, Scikit-Learn is used for Support Vector Machine (SVM) crop

classification, Principal Component Analysis (PCA) for feature visualization, and multiclass classification using the One-vs-One approach. Its optimized algorithms improve accuracy and simplify the machine learning workflow.

5.File Handling

Python file handling is used for reading, writing, and storing data in external files for further processing and hardware integration. In this project, file handling plays an important role in generating FPGA-compatible memory files that store processed image data, extracted feature vectors, SVM weights, and bias values. These text files are used as input memory files in Vivado simulation, enabling the trained Python-based machine learning model to be verified in an FPGA-oriented hardware environment.

6. Functions

Functions are reusable blocks of code that help organize complex programs into smaller, manageable sections for efficient execution. They improve code readability, modularity, debugging, and maintenance by avoiding repetition of similar operations. In this project, functions are used for tasks such as dataset loading, image preprocessing, hexadecimal data conversion, and PCA visualization, making the implementation more structured and easier to manage.

3.11 Verilog HDL

Verilog HDL (Hardware Description Language) is a programming language used to design and model digital hardware systems such as FPGA circuits. It allows designers to describe hardware behaviour, data flow, and logical operations using code, making simulation and verification possible before actual hardware implementation. In FPGA-based image processing systems, Verilog is used to implement arithmetic operations, memory handling, control logic, and processing modules. Important concepts such as modules, always blocks, registers, wires, sequential logic, and combinational logic help build efficient digital hardware architectures.

3.12 FPGA

FPGA (Field Programmable Gate Array) is a programmable digital hardware device widely used for high-speed and real-time processing applications. Unlike conventional software processors that execute instructions sequentially, FPGA performs multiple operations simultaneously through parallel hardware architecture, significantly improving processing speed and reducing latency. This makes FPGA highly suitable for computationally intensive tasks such as image processing, machine learning acceleration, and real-time signal analysis. Its reconfigurable nature also allows hardware designs to be modified and optimized based on application requirements.

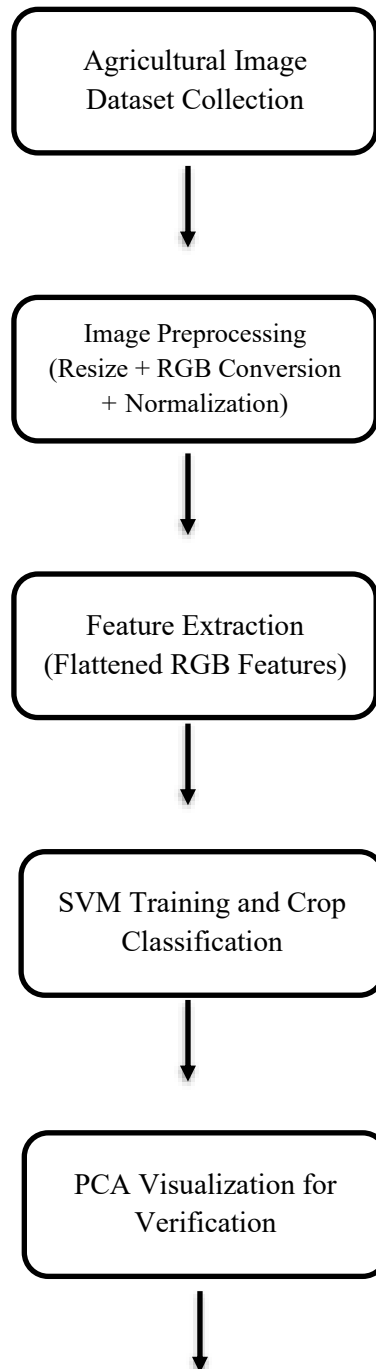
3.13 Vivado Simulation

Vivado is a comprehensive FPGA development tool used for designing, simulating, synthesizing, and verifying digital hardware systems. It provides an environment for writing Verilog HDL code, testing circuit functionality, analysing timing performance, and preparing designs for FPGA implementation. In FPGA-oriented image processing applications, Vivado simulation helps validate hardware behaviour and detect design errors before physical deployment, ensuring reliable and efficient system performance.

CHAPTER 4

METHODOLOGY

4.1 Overall System Workflow



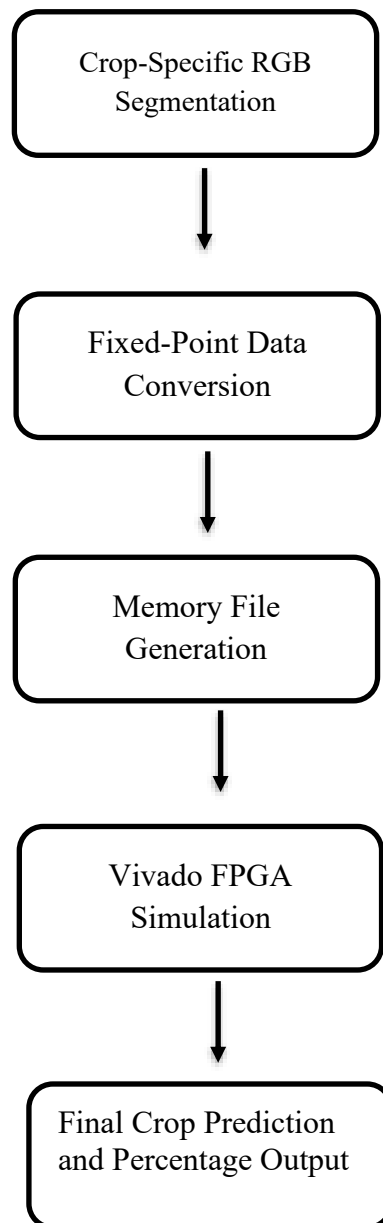


Fig. 4.1 Overall System Workflow

4.2 Agricultural Image Acquisition System

Agricultural monitoring systems require efficient image acquisition for automated crop analysis and field monitoring. In the proposed system, agricultural field images containing different crop types such as mirchi, cotton, paddy, and sunflower are collected and organized into a dataset for analysis. These images serve as the primary input for crop classification and crop percentage estimation.

Unlike manual agricultural monitoring, image-based analysis provides faster and more reliable crop observation. Digital images captured from agricultural fields can contain useful visual information such as crop color, texture, and spatial distribution. These characteristics help in identifying crop types and estimating crop coverage within the field.

The collected dataset is divided into crop-specific classes, enabling machine learning-based classification. Each image is resized to a fixed dimension to maintain uniformity and simplify further processing. Standardized image acquisition ensures compatibility with both software-based processing and FPGA-oriented implementation.

4.3 Image Preprocessing

Image preprocessing is an essential step in agricultural image analysis because raw images may vary in size, format, illumination, and colour distribution. Proper preprocessing improves classification accuracy and simplifies hardware implementation.

In the proposed system, all input images are resized to 64×64 pixels to create a fixed-size input representation. This standardization reduces computational complexity and ensures consistent feature dimensions for classification.

The input images are converted from BGR format to RGB format for easier colour-based crop analysis. RGB representation is particularly useful in agricultural applications because crop regions often exhibit distinct colour characteristics. For example, paddy fields generally show dominant green intensity, while cotton regions show brighter white colour distribution.

After conversion, pixel intensity values are normalized between 0 and 1 to improve machine learning performance and maintain numerical consistency. The RGB image is then flattened into a one-dimensional feature vector containing 12,288 features, which becomes the input for the classification model.

4.4 Crop Classification using Machine Learning

Crop classification is performed using a Support Vector Machine (SVM) classifier with a linear kernel. SVM is selected because of its simplicity, strong classification performance, and hardware-friendly mathematical structure suitable for FPGA implementation.

The classifier is trained using agricultural images belonging to four crop classes:

- Mirchi
- Cotton
- Paddy
- Sunflower

The flattened RGB feature vectors extracted during preprocessing are used as training inputs. Since multiple crop classes are involved, the One-vs-One (OvO) multiclass classification strategy is used.

In this method, separate binary classifiers are created for every pair of crop classes. For four crop categories, a total of six classifiers are generated:

- Mirchi vs Cotton
- Mirchi vs Paddy
- Mirchi vs Sunflower
- Cotton vs Paddy
- Cotton vs Sunflower
- Paddy vs Sunflower

Each classifier produces a decision value, and voting logic is used to determine the final crop prediction. This classification method provides efficient implementation for FPGA simulation because it mainly involves multiplication, addition, and comparison operations.

4.5 PCA-Based Visualization

To analyze the effectiveness of crop classification, Principal Component Analysis (PCA) is used for visualization. Agricultural image feature vectors are high-dimensional, making direct visualization difficult.

PCA reduces the dimensionality of the feature space into two principal components while preserving maximum data variation. The transformed data is plotted to visualize crop separation and classifier decision boundaries.

This step helps verify whether different crop categories are clearly distinguishable and confirms the performance of the trained SVM classifier.

4.6 Agricultural Field Segmentation and Crop Percentage Estimation

After crop classification, the predicted crop type is used for agricultural field segmentation. Segmentation separates crop regions from the background based on crop-specific RGB threshold conditions.

Different crops exhibit unique colour characteristics, allowing simple threshold-based segmentation.

The segmentation rules used are:

Mirchi: Red-dominant or green-dominant regions are selected.

Cotton: High-intensity white regions are selected.

Paddy: Green-dominant pixels are selected.

Sunflower: Yellowish regions with high red and green intensity are selected.

Binary masking is used to create segmented crop regions, where crop pixels are marked as white and non-crop pixels as black.

Crop percentage estimation is then calculated using:

$$\text{Crop Percentage} = \frac{\text{Number of Crop Pixels}}{\text{Total Number of Pixels}} \times 100$$

This provides an estimate of crop coverage within the agricultural image.

4.7 FPGA-Oriented Data Conversion

Since the proposed system follows an FPGA-oriented implementation approach, software-generated machine learning parameters must be converted into FPGA-compatible memory files.

The following data are exported:

- Test image RGB pixel values
- Feature vectors
- SVM weights
- SVM bias values

Floating-point values are converted into fixed-point integer representation using scaling for hardware compatibility.

Hexadecimal memory files are generated:

- test_image.txt
- feature_mem.txt
- svm_weights.txt
- svm_bias.txt

These files act as input memory for FPGA simulation.

4.8 FPGA Simulation using Vivado

To evaluate hardware feasibility, the trained crop classification system is simulated in Vivado using Verilog HDL.

The FPGA simulation performs:

- Feature loading from memory
- SVM multiplication and accumulation
- Bias addition
- One-vs-One voting logic
- Final crop prediction
- Crop percentage estimation

The SVM classifier is implemented using arithmetic hardware operations such as multiplication, accumulation, and comparison, making it suitable for FPGA-based parallel processing.

RGB pixel data are processed directly in hardware simulation to estimate crop percentage using threshold logic.

Although physical FPGA implementation is not performed, Vivado simulation validates the hardware-oriented design and confirms the feasibility of future real-time FPGA deployment.

CHAPTER 5

RESULT

5.1 Dataset Processing Results

The dataset was successfully loaded into the software environment for training and testing.

Software Output:

- Total Samples: 240
- Total Features Extracted: 12288
- Classification Algorithm: Multi-class SVM (One-vs-One)

This confirms that the dataset preprocessing and feature extraction stages were completed successfully.

5.2 PCA Visualization of Feature Distribution

Principal Component Analysis (PCA) was used to reduce the high-dimensional feature space into two principal components for visualization of crop class separability.

Observation:

The PCA plot shows distinct clustering of the four crop classes:

- Mirchi (Red points)
- Cotton (Gray points)
- Paddy (Green points)
- Sunflower (Yellow points)

The SVM decision boundaries clearly separate the classes, indicating effective feature extraction and successful classification capability.

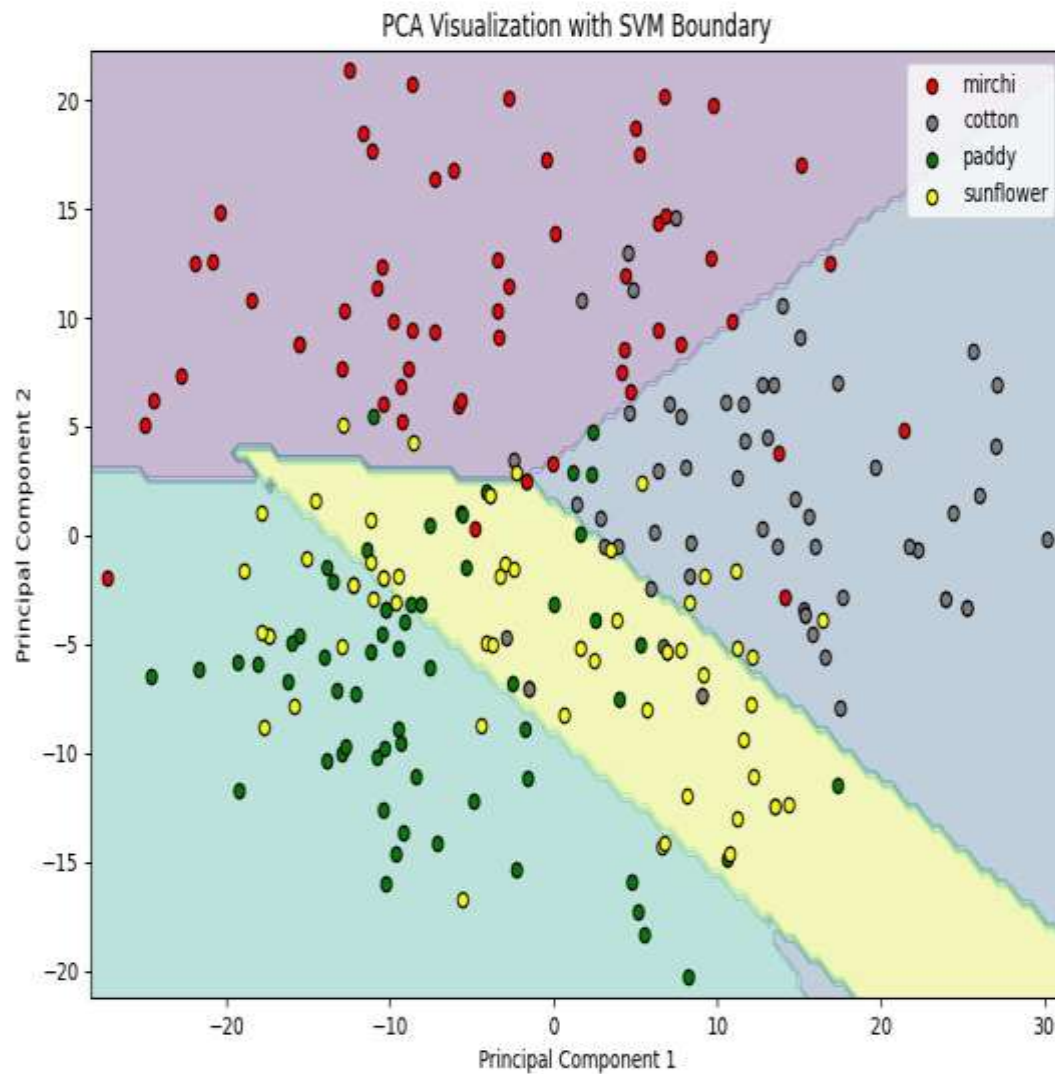


Fig. 5.1 PCA Visualization of Feature Distribution

5.3 Crop Classification and Yield Estimation Results

5.3.1 Mirchi

The input test image belonging to the Mirchi crop class was processed.

Software Prediction:

- Predicted Crop: **Mirchi**
- Decision Function Values:
[1.0002, 1.0342, 0.9996, 0.5818, 0.3712, -1.3857]

Binary Mask Result:

- Estimated Yield Percentage: **49.37%**

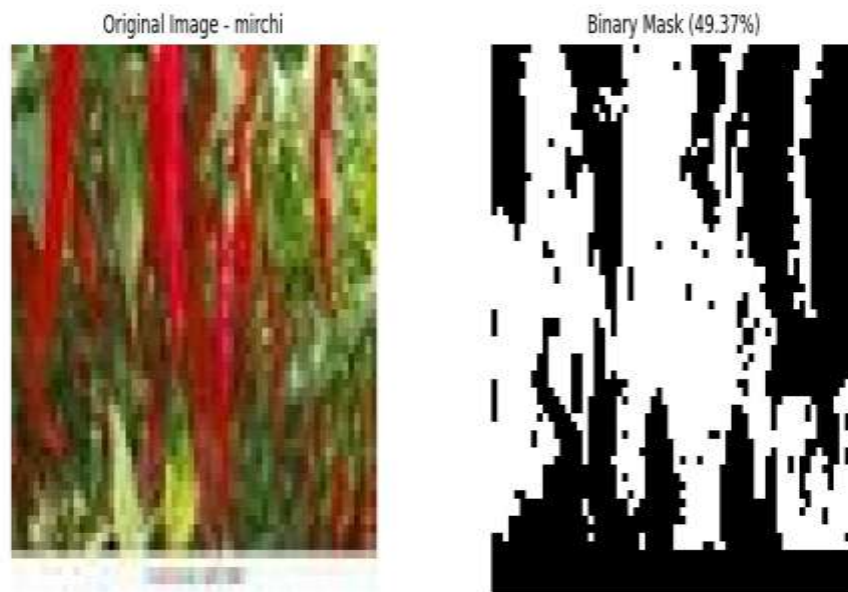


Fig.5.2.1 Original vs Binary Mask (Mirchi)

```
Loading Dataset...
Dataset Loaded - Samples: 240, Features: 12288

Training SVM (OvO)...
Training Completed
SOFTWARE PREDICTION
Predicted Crop : mirchi
Decision Function :
[ 1.00024755  1.03424764  0.99962845  0.5818602   0.37125004 -1.38576945]
Estimated Yield : 49.37%

Exporting Files...
FILES GENERATED SUCCESSFULLY
PS C:\Xilinx\Vivado\2024.2>
```

Fig. 5.2.2 Python Simulation Output (Mirchi)

FPGA Hardware Verification:

Vivado simulation and hardware outputs confirm:

- Winner Class = **0 (Mirchi)**
- Yield Count = **2022**
- Yield Percentage = **49%**



Fig. 5.2.3 Vivado Simulation (Mirchi)

Name	Value
clk	1
reset	0
start	1
done	1
> feature_index[31:0]	12288
> feature_mem[0:12287][31:0]	00000e3f,00000fa0,00000781,00000f03,0000119d,00000930,00000fc7,0000154a,00000b7
weight_mem[0:73727][31:0]	fffffffd,fffffff,fffffff4,fffffffe,ffffff,ffffff4,ffffffd,ffffff,ffffff3,ffffffb,ffffffd,ffffff0,ffff
> bias_mem[0:5][31:0]	00000975,ffffffce1,00000382,ffffd85d,ffffd9ae,00000373
> image_mem[0:4095][23:0]	5d6631,62733c,678b4b,abce8e,434c15,928455,b1875f,86351a,9a0d06,ad0407,b90f12,9d
> acc[0:5][95:0]	0000000000000000004774a07,00000000000000000670e332,000000000000000005590e0;
> votes[0:3][2:0]	3,2,0,1
> i[31:0]	4096
> c[31:0]	6
> idx[31:0]	73727
> winner[31:0]	0
> yield_count[31:0]	2022
> yield_percentage[31:0]	49
done_prev	0
> r[7:0]	eb
> g[7:0]	ff
> b[7:0]	fd

Fig. 5.2.4 Simulation outputs (Mirchi)

5.3.2 Cotton

The Cotton crop image was processed.

Software Prediction:

- Predicted Crop: **Cotton**
- Decision Function:
[-0.9998, 0.0259, 0.2356, 1.0956, 1.3576, -0.6782]

Binary Mask Yield:

- Estimated Yield = **44.63%**

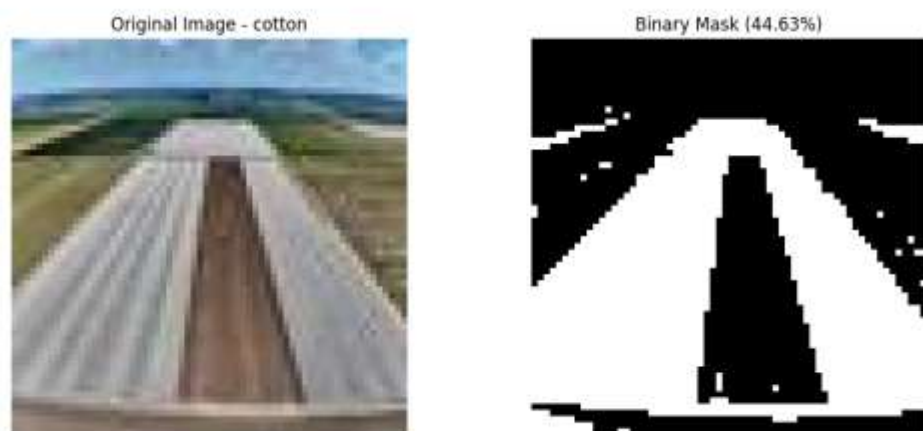


Fig.5.3.1 Original vs Binary Mask (Cotton)

```
Loading Dataset...
Dataset Loaded - Samples: 240, Features: 12288

Training SVM (OvO)...
Training Completed
SOFTWARE PREDICTION
Predicted Crop : cotton
Decision Function :
[-0.99981088  0.02593091  0.23569336  1.09566907  1.35763197 -0.67826754]
Estimated Yield : 44.63%

Exporting Files...
FILES GENERATED SUCCESSFULLY
PS C:\Xilinx\Vivado\2024.2> 
```

Fig. 5.3.2 Python Simulation Output (Cotton)

FPGA Verification:

- Winner Class = **1 (Cotton)**
- Yield Count = **1828**
- Yield Percentage = **44%**



Fig. 5.3.3 Vivado Simulation (Cotton)

Name	Value
clk	1
reset	0
start	1
done	1
feature_index[31:0]	12288
feature_mem[0:12287][31:0]	0000108b,00001882,00001e2d,0000128
weight_mem[0:73727][31:0]	ffffffffd,ffffffff,ffffffff4,fffffffe,ffffff,ffffff
bias_mem[0:5][31:0]	00000975,ffffce1,00000382,ffffd85d,ffff
image_mem[0:4095][23:0]	6ca0c5,79abd0,8dbbdf,9fc9ef,add0f6,ac
acc[0:5][95:0]	ffffffffffff8c5995c,00000000000000000000c
votes[0:3][2:0]	2,3,0,1
i[31:0]	4096
c[31:0]	6
idx[31:0]	73727
winner[31:0]	1
yield_count[31:0]	1828
yield_percentage[31:0]	44
done_prev	0
r[7:0]	74
g[7:0]	68
b[7:0]	5c

Fig. 5.3.4 Simulation outputs (Cotton)

5.3.4 Paddy

The Paddy crop image was tested.

Software Prediction:

- Predicted Crop: **Paddy**
- Decision Function:
[-0.7034, -1.4595, -2.0382, -1.4372, -1.5382, 1.0001]

Binary Mask Result:

- Estimated Yield = **83.33%**

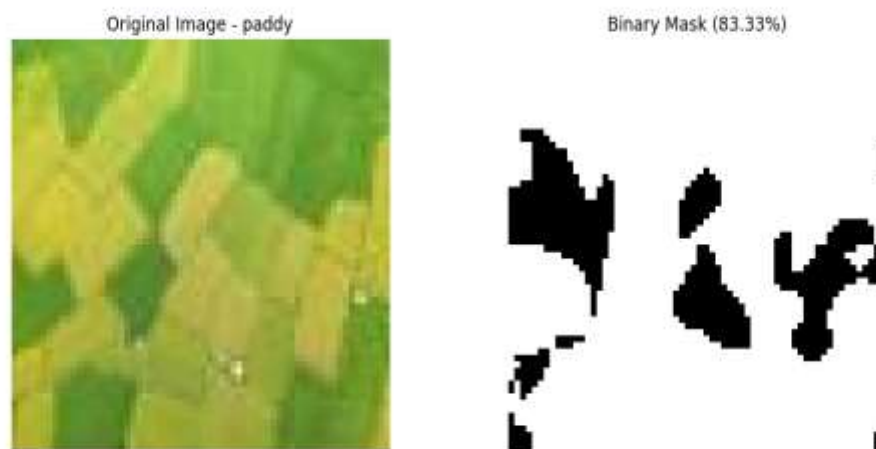


Fig.5.4.1 Original vs Binary Mask (Paddy)

```
Loading Dataset...
Dataset Loaded - Samples: 240, Features: 12288

Training SVM (OvO)...
Training Completed
SOFTWARE PREDICTION
Predicted Crop : paddy
Decision Function :
[-0.70344241 -1.45951663 -2.0382225 -1.43724349 -1.5382059 1.00012218]
Estimated Yield : 83.33%

Exporting Files...
FILES GENERATED SUCCESSFULLY
PS C:\Xilinx\Vivado\2024.2> 
```

Fig. 5.4.2 Python Simulation Output (Paddy)

FPGA Verification:

- Winner Class = **2 (Paddy)**
- Yield Count = **3413**
- Yield Percentage = **83%**



Fig. 5.4.3 Vivado Simulation (Paddy)

Name	Value
clk	1
reset	0
start	1
done	1
feature_index[31:0]	12288
feature_mem[0:12287][31:0]	00000dc9,000017e5,000008e2,00000fee,000019e3,00000a43,00000fc7,000018d0
weight_mem[0:73727][31:0]	ffffffff,ffffffff,ffffff4,ffffffe,ffffff,ffffff4,ffffffd,ffffff,ffffff3,ffffffb,ffffffd,fffff
bias_mem[0:5][31:0]	00000975,ffffce1,00000382,ffffd85d,ffffd9ae,00000373
image_mem[0:4095][23:0]	5a9c3a,68a943,67a238,70a938,629922,70a62c,6aa225,6faa2c,6fad32,6faf34,71b2
acc[0:5][95:0]	fffffffffffffffffa7f67f9,ffffffffffffffff7e651ad,ffffffffffffffff3892782,ffffffffffffffffd8
votes[0:3][2:0]	0,1,3,2
i[31:0]	4096
c[31:0]	6
idx[31:0]	73727
winner[31:0]	2
yield_count[31:0]	3413
yield_percentage[31:0]	83
done_prev	0
r[7:0]	b6
g[7:0]	af
b[7:0]	3d

Fig. 5.4.4 Simulation outputs (Paddy)

5.3.5 Sunflower

The Sunflower crop image was processed.

Software Prediction:

- Predicted Crop: **Sunflower**
- Estimated Yield = **66.02%**

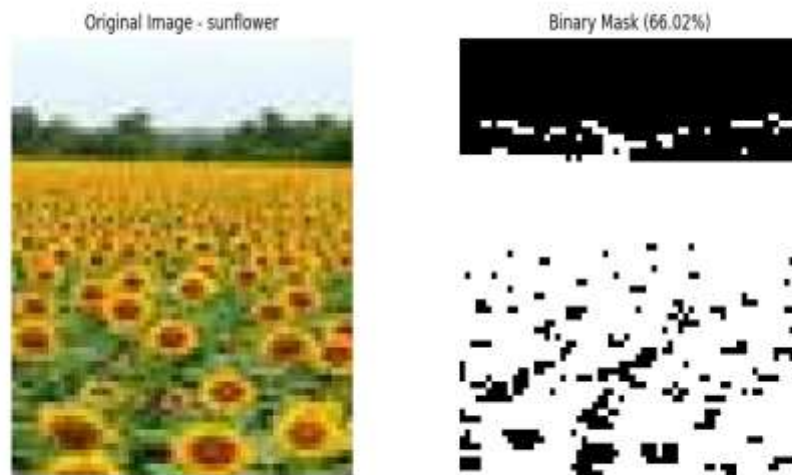


Fig.5.5.1 Original vs Binary Mask (Sunflower)

```
Loading Dataset...
Dataset Loaded - Samples: 240, Features: 12288

Training SVM (OvO)...
Training Completed
SOFTWARE PREDICTION
Predicted Crop : sunflower
Decision Function :
[ 0.05936363 -0.67496377 -1.52767743 -0.55918729 -1.66804652 -0.99975973]
Estimated Yield : 66.02%

Exporting Files...
FILES GENERATED SUCCESSFULLY
PS C:\Xilinx\Vivado\2024.2> █
```

Fig. 5.5.2 Python Simulation Output (Sunflower)

FPGA Verification:

- Winner Class = **3 (Sunflower)**
- Yield Percentage = **66%**

Name	Value
clk	1
reset	0
start	1
done	1
> feature_index[31:0]	00003000
> bias_mem[0:5][31:0]	00000975,ffffce1,00000382,ffffd85d,ffffd9ae,00000373
> acc[0:5][95:0]	ffffffffffffe18687d,ffffffffffffc742f7b,fffffffffffff810da,000000000...
> votes[0:3][2:0]	0,0,1,3
> i[31:0]	00001000
> c[31:0]	00000006
> idx[31:0]	00011fff
> winner[31:0]	00000003
> yield_count[31:0]	00000a90
> yield_percentage[31:0]	00000042

Fig. 5.5.3 Vivado Simulation (Sunflower)

Name	Value
clk	1
reset	0
start	1
done	1
> feature_index[31:0]	12288
> feature_mem[0:12287][31:0]	0000249c,000026e8,000025d6,0000249c,000026e8,000025d6,00
> weight_mem[0:73727][31:0]	fffffffd,fffffff4,fffffff4,fffffff4,fffffff4,fffffff4,fffffff4,fffff
> bias_mem[0:5][31:0]	00000975,fffcc1,00000382,fffd85d,fffd9ae,00000373
> image_mem[0:4095][23:0]	effer7,effer7,effef9,effeb,effdf,effdf,effdf,effdf,edfbfe,edbf
> acc[0:5][95:0]	ffffffffffffffffffef8687d,ffffffffffffffffffc742f7b,ffffffffffffffffff6810d0,
> votes[0:3][2:0]	0,2,1,3
> i[31:0]	4096
> c[31:0]	6
> idx[31:0]	73727
> winner[31:0]	3
> yield_count[31:0]	2704
> yield_percentage[31:0]	66
done_prev	0
> r[7:0]	66
> g[7:0]	79
> b[7:0]	4b

Fig. 5.5.4 Simulation outputs (Sunflower)

5.4 Software vs FPGA Comparison

Crop	Software Prediction	FPGA Prediction	Software Yield (%)	FPGA Yield (%)
Mirchi	Mirchi	Mirchi	49.37	49
Cotton	Cotton	Cotton	44.63	44
Paddy	Paddy	Paddy	83.33	83
Sunflower	Sunflower	Sunflower	66.02	66

Fig. 5.6 Software vs FPGA Comparison

5.5 RTL Schematic of FPGA-Based Crop Detection and Crop Percentage Estimation System

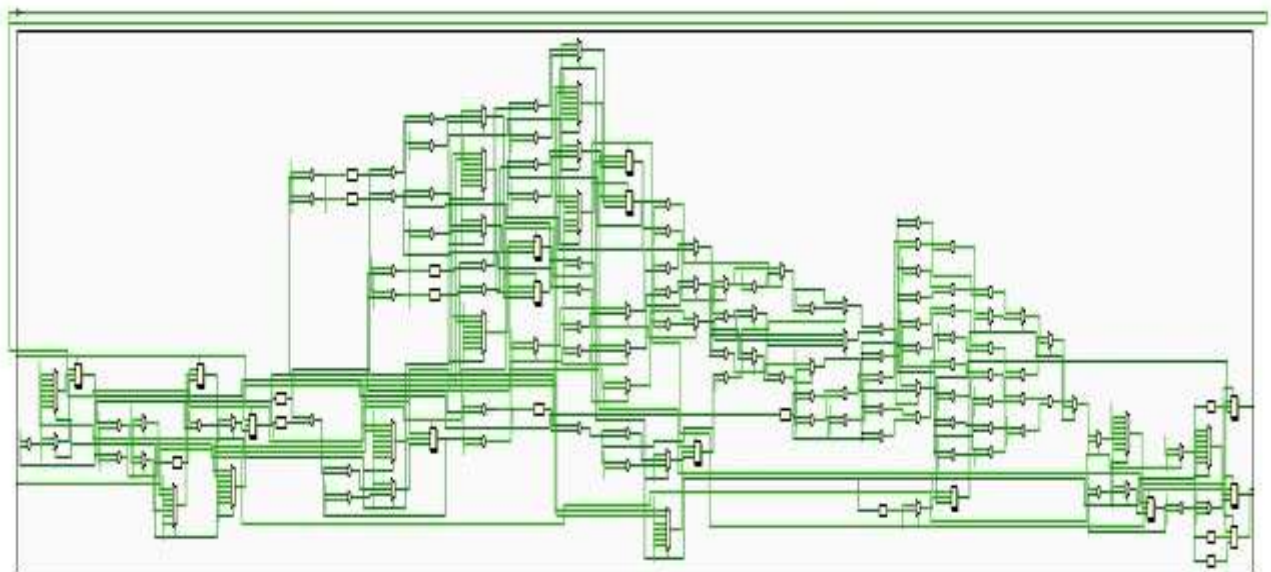


Fig.5.7 RTL Schematic of FPGA-Based Crop Detection and Crop Percentage Estimation System

CONCLUSION

The proposed project, FPGA-Based Multi-Crop Classification and Yield Estimation using Image Processing and SVM, was successfully implemented and tested. The system automatically identifies crop types and estimates yield from input images. It was trained to classify Mirchi, Cotton, Paddy, and Sunflower using image preprocessing, feature extraction, and SVM classification, with PCA showing clear separation between crop classes for effective classification.

The experimental results showed accurate crop prediction with yield estimations of Mirchi – 49.37%, Cotton – 44.63%, Paddy – 83.33%, and Sunflower – 66.02%. Binary masking was used for yield estimation by calculating crop coverage from the processed images. The trained model was successfully implemented on FPGA using Xilinx Vivado, and the hardware outputs closely matched the software results, confirming correct and reliable hardware operation.

The project also demonstrates the capability of FPGA for real-time processing, faster computation, and efficient hardware implementation compared to software-only systems. Overall, the combination of image processing, machine learning, and FPGA technology provides an effective solution for crop classification and yield estimation, making it suitable for precision agriculture, smart farming, and real-time crop monitoring applications. Future improvements can include adding more crop types, larger datasets, and advanced AI techniques for higher accuracy.

FUTURE SCOPE

- The proposed system can be extended to classify more crop varieties by increasing the dataset and training the model with additional crop images.
- The classification accuracy can be improved by using advanced machine learning and deep learning algorithms such as CNNs, ResNet, or hybrid AI models instead of basic SVM classification.
- The project can be integrated with IoT sensors to collect real-time environmental parameters such as temperature, humidity, soil moisture, and light intensity for better crop analysis.
- The system can be connected with drone or satellite imaging to monitor large agricultural fields and perform large-scale crop classification and yield estimation.
- FPGA implementation can be optimized further to achieve higher processing speed, lower power consumption, and better resource utilization.
- A real-time embedded system can be developed by integrating the FPGA with cameras for live crop monitoring directly in agricultural fields.
- The project can be enhanced to detect crop diseases, pest infections, and nutrient deficiencies along with crop classification.

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